MATHEMATICS SECTION 1 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR options** (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks : +3 If **ONLY** the correct option is chosen
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.1 Let $\mathbb{R}$ denote the set of all real numbers. Then the value of $\sum_{x \in S}(x^2 + x\cos^{-1}x)$ for the set

$$S = \{x \in \mathbb{R} : \tan^{-1}(x-1) + \tan^{-1}(x+1) = \tan^{-1}(3x) - \tan^{-1}(x)\}$$

is

(A) $\frac{1}{2} - \frac{\pi}{6}$	**(B)** $\frac{1}{4} - \frac{\pi}{3}$
(C) $\frac{3}{4} - \frac{\pi}{6}$	**(D)** $\frac{1}{2} + \frac{\pi}{3}$

Answer: A

Q.2 Let $P(x) = x^3 + x^2 + x - 1$ and λ be its unique real root. A function $f : \mathbb{R} \to \mathbb{R}$ is defined as

$$f(x) = \frac{kx^3 + (k-1)x^2 + (k-2)x + (3-k)}{3x^2 + 2x + 1}$$

where k is a real parameter. If $L = \lim_{x \to \lambda} \frac{f(x) - \lambda}{x - \lambda}$, then which of the following statements is TRUE?

(A) If $k = 1$, then $L = -2$	**(B)** If $k = 2$, then $L = 1$
(C) If $k = 4$, then $L = 4$	**(D)** If $k = 5$, then $L = 3$

Answer: D

Q.3 Let P be a point in the first quadrant on the parabola $y^2 = 12x$. Let the normal to the parabola at P intersect the x-axis at point N. Let T be the point of intersection of the tangent to the parabola at P and the line passing through N and parallel to the y-axis. If the area of the triangle PNT is 120, then the radius of the circumcircle of the triangle PNT is

(A) 10	**(B)** 20
(C) 30	**(D)** 40

Answer: B

Q.4

Let $M_n(\mathbb{R})$ denote the set of all $n \times n$ matrices with real entries. Define a matrix $A \in M_3(\mathbb{R})$ such that $A \neq 0$ but $A^3 = 0$. Let I be the 3×3 identity matrix. Define $f(t) = \det(I + tA)$ for any real number t.

Then which one of the following statements is TRUE?

(A) The function $f(t)$ is a strictly increasing function of t for all $t > 0$.	**(B)** There exists a non-zero matrix $B \in M_3(\mathbb{R})$ such that $AB - BA = I$.
(C) For any non-zero real number α, the matrix $(I - \alpha A)^{-1}$ exists and is equal to $I + \alpha A + \alpha^2 A^2$.	**(D)** The trace of matrix A is strictly greater than 0.

Answer: C

MATHEMATICS SECTION 2 (Maximum Marks : 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

Q.5 Let $f : [0, \infty) \to \mathbb{R}$ be a differentiable function such that $f(0) = 0$ and

$$f'(x) = \frac{e^{-x}}{1 + (f(x))^2}$$

for all $x \geq 0$. Then which of the following statements is (are) TRUE?

(A) The equation of the tangent to the curve $y = f(x)$ at $x = 0$ is $y = x$	**(B)** The curve $y = f(x)$ has at least one point of inflection in the interval $(0, \infty)$
(C) $\lim_{x\to\infty} f(x)$ exists and its value lies in the interval $\left(\frac{4}{5}, 1\right)$	**(D)** $f(x) > \frac{3}{4}(1 - e^{-x})$ for all $x > 0$

Answer: A, C, D

Q.6 Let $n \in \mathbb{N}$. Define the sequences a_n and b_n as follows:

$$a_n = \sum_{k=0}^{n} \binom{n}{k}^2$$

$$b_n = \sum_{k=0}^{n} \binom{n}{k} \cos\left(\frac{k\pi}{3}\right)$$

Let $T_1 = \{a_n : n \in \mathbb{N}\}$ and $T_2 = \{b_n : n \in \mathbb{N}\}$.

Which of the following statements is (are) TRUE?

(A) $T_2 \cap (-\infty, 0) = \phi$, where ϕ denotes the empty set.	**(B)** For all $n \in \mathbb{N}$, $a_n = \frac{2}{\pi} \int_0^{\pi/2} (2\cos x)^{2n} dx$
(C) $\lim_{n\to\infty} \frac{\ln(a_n)}{n} = \ln 4$	**(D)** If $c_n = \sum_{k=0}^{n} \binom{n}{k} \sin\left(\frac{k\pi}{3}\right)$, then $b_n^2 + c_n^2 = 3^n$ for all $n \in \mathbb{N}$.

Answer: B, C, D

Q.7

Let two sequences of real numbers, $S = \{x_1, x_2, x_3, \ldots\}$ and $T = \{y_1, y_2, y_3, \ldots\}$, be defined as follows:

For all natural numbers n, the nth term of sequence S is:

$$x_n = \sum_{k=1}^{n} \frac{2k+1}{(k^2+k)^2}$$

Sequence T is defined by $y_1 = 1$, and for $n \geq 2$, it satisfies the recurrence relation:

$$ny_n - (n-1)y_{n-1} = 2n - 1$$

Which of the following statements is (are) TRUE?

(A) The infinite series $\sum_{n=1}^{\infty}(1 - x_n)$ converges to an irrational number.	**(B)** $\lim_{n\to\infty} \prod_{k=1}^{n} x_k = \frac{1}{2}$
(C) For any integer $n \geq 2$, the sum of the cubes of the first n terms of T is strictly greater than the square of the sum of the first n terms of T.	**(D)** $\lim_{n\to\infty} \left(y_n^2 \cdot (1 - x_{y_n})\right) = 1$

Answer: A, B, D

Q.8

Let S denote the locus of the foot of the perpendicular drawn from the center of the hyperbola $\frac{x^2}{16} - \frac{y^2}{9} = 1$ to any of its tangent lines. Let R denote the region enclosed by a single loop of the curve S lying in the right half-plane ($x \geq 0$).

Then which of the following statements is (are) TRUE?

(A) The equations of the tangents to the curve S at the origin are $3x \pm 4y = 0$	**(B)** The maximum distance of any point on the curve S from the origin is 4
(C) The area of the region R is $6 + \frac{7}{2}\sin^{-1}\left(\frac{4}{5}\right)$	**(D)** The area of the region R is $12 + \frac{7}{2}\cos^{-1}\left(\frac{3}{5}\right)$

Answer: B, C

MATHEMATICS SECTION 3 (Maximum Marks : 20)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 If **ONLY** the numerical answer is in acceptable range
  Zero Marks : 0 If no numerical answer is given (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases

Q.9 Let A and d be positive real numbers. For each $n \in \mathbb{N}$, let α_n and β_n ($\alpha_n < \beta_n$) be the real roots of the quadratic equation:

$$x^2 - \left(\int_0^1 (2nd + d)dt\right) x + d^2 \left(\int_0^n (2t+1)dt\right) = 0$$

Let I_n be defined by the definite integral:

$$I_n = \int_{\alpha_n}^{\beta_n} \frac{A}{x(x+d)} dx$$

If $\int_{\alpha_1}^{\beta_1} t\, dt = 24$, and

$$\sum_{n=1}^{30} I_n = 24 \ln\left(\frac{31}{16}\right),$$

then the value of λ in the equation

$$\sum_{n=3}^{14} I_n = \frac{\lambda}{4} \ln\left(\frac{5}{4}\right)$$

is ____.

Answer: 96

Q.10

Let $u = \dfrac{\sqrt{3}\hat{i} + \hat{j}}{2}$ and $v = \dfrac{-\hat{i} + \sqrt{3}\hat{j} + 2\sqrt{3}\hat{k}}{4}$.

Let d be the shortest distance between the line $L : \dfrac{x}{2} = \dfrac{y}{-4} = \dfrac{z}{1}$ and the plane $\Pi : x + y + 2z = 6$.

If $x, y \in \mathbb{R}$ are such that:

$$3x + 2y = 10\left(|u \times v|^2 + (u \cdot v)^2\right)^{\frac{5}{4}}$$

and

$$x - 2y = d^2$$

then $2x + 3y$ is equal to ____.

Answer: 5

Q.11

Let $P(x)$ be a polynomial of degree 3 such that $P(1) = 1, P(2) = 2$, and $P(3) = 3$.

Suppose that the value of the definite integral $\int_0^1 P(x)dx = -4$. A sequence $\{t_n\}$ is defined for $n \geq 4$ as:

$$t_n = \frac{1}{P(n) - n}$$

If the sum of the infinite series $S = \sum_{n=4}^{\infty} t_n$, then the value of $120S$ is ____.

Answer: 15

Q.12

Let the function $f : \mathbb{R} \to \mathbb{R}$ be defined by $f(x) = x^5 + 2x^3 + 3x$. Suppose $g(x)$ denotes the uniquely defined inverse function of $f(x)$. Define the definite integral $\alpha = \int_0^6 g(x)dx$ and let the limit $\beta = \lim_{x \to 0} \dfrac{1}{x^3} \int_0^x (x - t)^2 e^{t^2} dt$. Then the value of $\alpha + \beta$, rounded off to two decimal places, is equal to ____.

Answer: [4.1 to 4.2]

Q.13

Let $S = \{1, 2, 3, 4, 5, 6\}$. Let $g : S \to S$ be a fixed function defined as $g(1) = g(2) = g(3) = 1, g(4) = g(5) = 2$, and $g(6) = 3$.

Find the number of functions $f : S \to S$ such that:

1. $f(f(x)) = x$ for all $x \in S$
2. $g(f(x)) = g(x)$ for all $x \in S$

Answer: 8

MATHEMATICS SECTION 4 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +3 If **ONLY** the numerical answer is in acceptable range
  Zero Marks : 0 In all other cases

Q.14

Paragraph: Let $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$. For any subset A of S, define its "Binomial Weight" as $W(A) = \sum_{x \in A} {}^7C_x$. (If A is the null set, $W(A) = 0$).

Let X be the collection of all subsets A of S that satisfy the following property:
i. $W(A)$ is an odd integer.

Let $Y = \{A \in X : n(A) = 4\}$ and
$Z = \{A \in X : \text{The maximum element in } A \text{ is } 5\}$.

(Note: $n(A)$ denotes the number of elements in a set A.)

Question: If $\log_4(n(X)) = k$, then the value of k is ____.

Answer: 3.5

Q.15

Paragraph: Let $S = \{0, 1, 2, 3, 4, 5, 6, 7\}$. For any subset A of S, define its "Binomial Weight" as $W(A) = \sum_{x \in A} {}^7C_x$. (If A is the null set, $W(A) = 0$).

Let X be the collection of all subsets A of S that satisfy the following property:
i. $W(A)$ is an odd integer.

Let $Y = \{A \in X : n(A) = 4\}$ and
$Z = \{A \in X : \text{The maximum element in } A \text{ is } 5\}$.

(Note: $n(A)$ denotes the number of elements in a set A.)

Question: If the value of $n(Y) + n(Z) = p^2$ (where $p > 0$), then the value of p is ____.

Answer: 4

Q.16

Paragraph: Let $f:[0,3]\rightarrow\mathbb{R}$ be the function defined by $f(x)=\dfrac{x^2}{x^2+(3-x)^2}$ and let $g:[0,3]\rightarrow\mathbb{R}$ be the function defined by $g(x)=e^{x(3-x)}\sin\left(\dfrac{\pi x}{3}\right)$.

The answer to the question is a NUMERICAL VALUE. If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.

Question: The value of $\int_0^3 f(x)g(x)dx-\dfrac{1}{2}\int_0^3 g(x)dx$ is ____.

Answer: 0

Q.17

Paragraph: Let $f:[0,3]\rightarrow\mathbb{R}$ be the function defined by $f(x)=\dfrac{x^2}{x^2+(3-x)^2}$ and let $g:[0,3]\rightarrow\mathbb{R}$ be the function defined by $g(x)=e^{x(3-x)}\sin\left(\dfrac{\pi x}{3}\right)$.

The answer to the question is a NUMERICAL VALUE. If the numerical value has more than two decimal places, truncate/round-off the value to TWO decimal places.

Question: The value of $\int_0^3 x(3-x)f(x)dx$ is ____.

Answer: 2.25

PHYSICS SECTION 1 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR options** (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

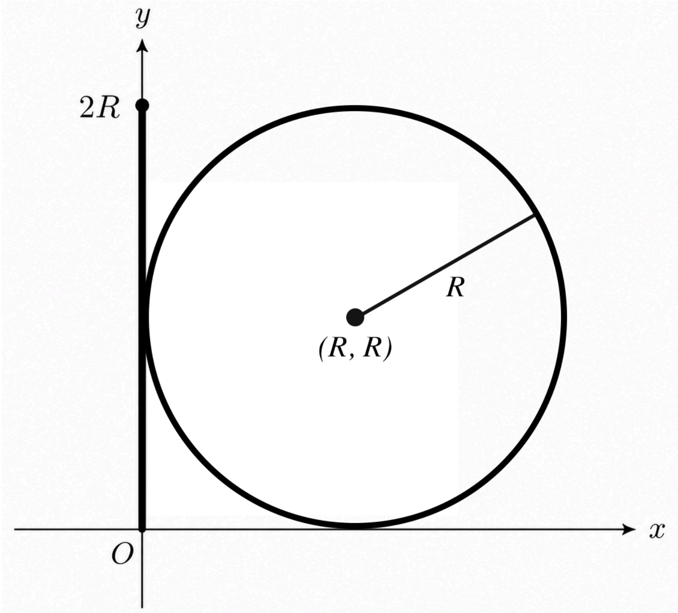
  Full Marks : +3 If **ONLY** the correct option is chosen
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases.

Q.18 A Vernier caliper (where N vernier scale divisions coincide with $N-1$ main scale divisions) is made of a metal with a linear expansion coefficient α_1. It is used to measure the side of a cube (linear expansion coefficient α_2). If the caliper reads L_0 at temperature T, and L' at $T+\Delta T$, then for a small ΔT, the quantity $Q = \dfrac{L' - L_0}{L_0 \Delta T}$ is:

(A) $\alpha_2 - \alpha_1$	**(B)** $\alpha_1 + \alpha_2$
(C) $\alpha_2 - \alpha_1\left(\dfrac{N-1}{N}\right)$	**(D)** $\alpha_1 - \alpha_2$

Answer: A

Q.19



A rigid planar structure is constructed in the xy-plane by combining a straight thin rod and a thin circular ring.

The straight thin rod of length $2R$ lies entirely along the positive y-axis. Its bottom end is at the origin $(0, 0)$ and its top end is at $(0, 2R)$. The linear mass density λ of this rod varies with the distance y from the origin according to the relation $\lambda(y) = \lambda_0\left(\frac{y}{2R}\right)$, where λ_0 is a positive constant. Let the total mass of this rod be M_{rod}.

The uniform thin circular ring of radius R is attached to the rod. The ring lies entirely in the xy-plane and is centered at the coordinates (R, R). The total mass of this uniform ring is given to be exactly $2M_{rod}$.

Find the total moment of inertia of this combined structure about the x-axis.

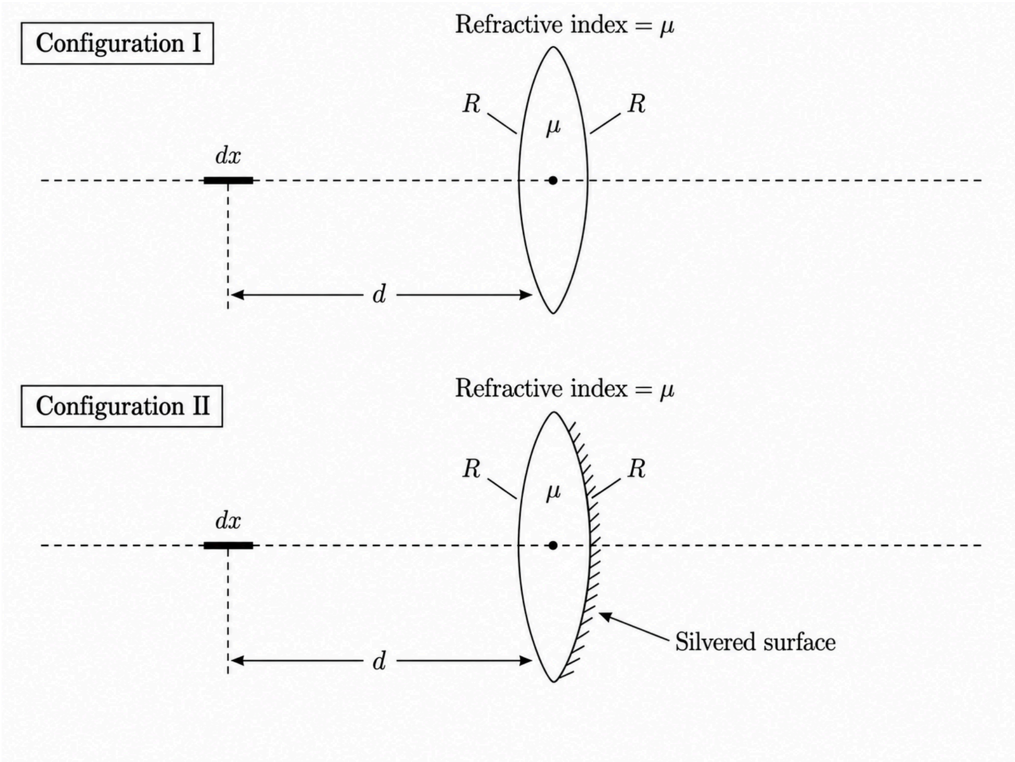
(A) $3M_{rod}R^2$	**(B)** $4M_{rod}R^2$
(C) $5M_{rod}R^2$	**(D)** $6M_{rod}R^2$

Answer: C

Q.20 A small linear object of length dx (where $dx \ll R$) lies along the principal axis of a thin equiconvex lens. The lens is made of a material with refractive index μ and has radii of curvature R for both of its spherical surfaces. The center of the object is located at a distance d (where $d > R$) from the optical center of the lens. Let the length of the image formed by refraction through this lens be l_1.

In a second configuration, the back surface of the exact same lens is silvered. The same small object is placed at the exact same distance d along the principal axis in front of the unsilvered surface. Let the length of the image formed by this silvered lens system be l_2.

Assuming paraxial rays and that the object is small enough that the longitudinal magnification can be approximated using the transverse magnification ($m_L \approx m_T^2$), what is the ratio $\dfrac{l_1}{l_2}$?



(A) $\left[\dfrac{R-2d(\mu-1)}{R-2d(2\mu-1)}\right]^2$	**(B)** $\dfrac{R-2d(2\mu-1)}{R-2d(\mu-1)}$
(C) $\left[\dfrac{R-d(2\mu-1)}{R-2d(\mu-1)}\right]^2$	**(D)** $\left[\dfrac{R-2d(2\mu-1)}{R-2d(\mu-1)}\right]^2$

Answer: D

Q.21 A motorboat crosses a river of width d strictly along the shortest possible path. The boat's speed in still water is a constant v, and the river's current speed is u (where $v > u$). The current speed u slowly decreases at a constant positive rate α. The boat continuously adjusts its heading to maintain this shortest path.

Let T be the instantaneous predicted crossing time, defined as the time the boat would take to cross the river if the current speed u at any given instant were to remain constant throughout the journey. What is the relative rate of change of this crossing time, $\frac{1}{T}\frac{dT}{dt}$?

(A) $\frac{\alpha u}{v^2 - u^2}$	**(B)** $-\frac{\alpha u}{v^2 - u^2}$
(C) $\frac{\alpha}{2(v^2 - u^2)}$	**(D)** $-\frac{\alpha}{v}$

Answer: B

PHYSICS SECTION 2 (Maximum Marks : 16)

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- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
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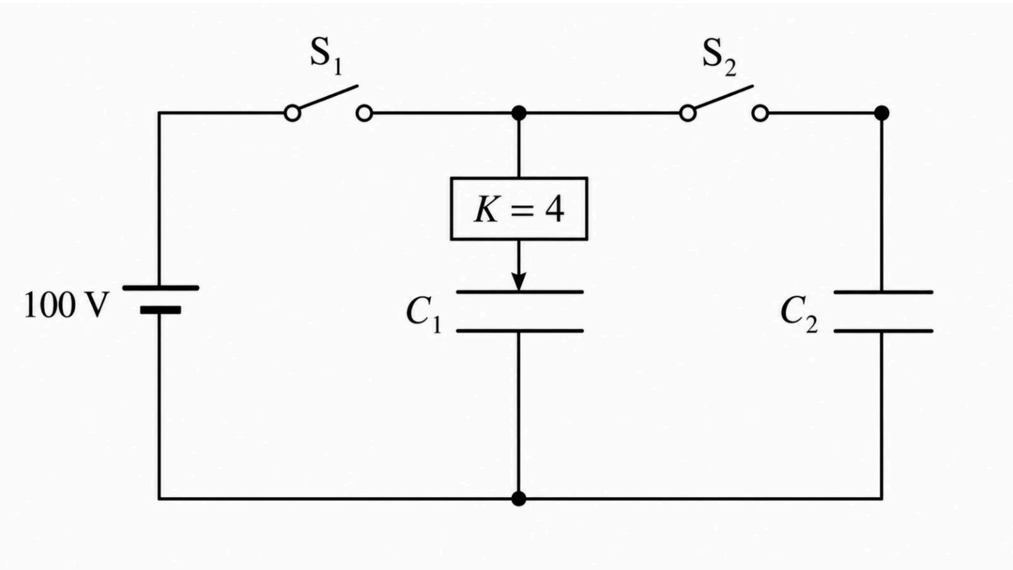
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

Q.22 An air-filled parallel plate capacitor of capacitance 20×10^{-6} F is fully charged by a 100 V battery. The battery is then disconnected and the charge on the capacitor is measured. Subsequently, a dielectric slab of dielectric constant $K = 4$ is completely inserted between its plates. Finally, this capacitor is connected in parallel to a second, identical uncharged air-filled capacitor. Taking the initial energy stored in the first capacitor as U_0, which of the following statements is/are correct:



(A) The electrostatic potential difference across the first capacitor immediately after inserting the dielectric is 25 V.	**(B)** The charge flowing to the second capacitor upon final connection is 1.6×10^{-3} C.
(C) The ratio of the final total energy of the system to the initial energy U_0 is $1 : 5$.	**(D)** The final electric field between the plates of the first capacitor is reduced to one-fifth of its initial value.

Answer: A, C, D

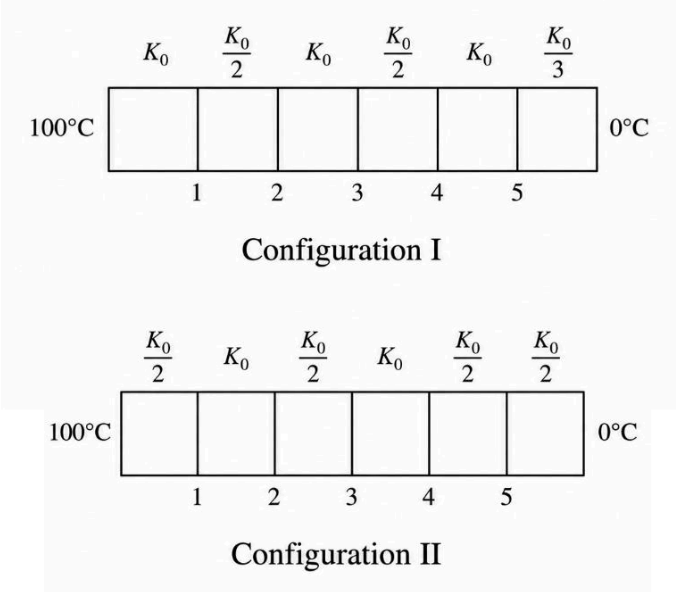
Q.23 A particle of mass m ($m \neq M$) moving with an initial speed v is absorbed by a stationary atom of mass M, forming a single combined system. Subsequently, the combined system emits a single photon of energy ΔE along the initial line of motion. Assume the combined system possesses sufficient internal energy to account for the photon's emission, and treat the motion of all masses non-relativistically ($v \ll c$).

If the final speed of the combined system after emitting the photon is exactly half the initial speed of the incident particle, which of the following is/are possible expressions for the initial kinetic energy of the particle? (Let c be the speed of light in vacuum).

(A) $\dfrac{2m(\Delta E)^2}{c^2(m+M)^2}$	**(B)** $\dfrac{2m(\Delta E)^2}{c^2(m+3M)^2}$
(C) $\dfrac{2m(\Delta E)^2}{c^2(m-M)^2}$	**(D)** $\dfrac{2m(\Delta E)^2}{c^2(3m+M)^2}$

Answer: C, D

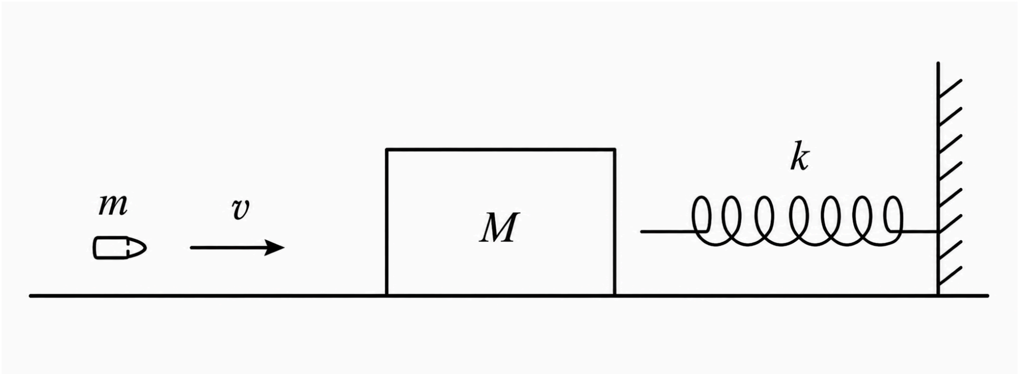
Q.24 Six uniform, solid slabs of equal cross-sectional area are joined face-to-face in series in configurations I and II. As shown in the figure, the slabs have uniform thermal conductivities indicated in terms of K_0. The thickness of each slab is 1 cm. The various junction interfaces between the slabs are denoted as $1, 2, 3, 4,$ and 5. The extreme left and right exposed faces are maintained at $100°$C and $0°$C respectively. Steady-state heat conduction is achieved. If $K_0 = 200$ W/(m·K), then which of the following statements is/are correct:



(A) At junction 4 of configuration I, the temperature is $40°$C.	**(B)** At junction 3 of configuration II, the temperature is $50°$C.
(C) The temperature difference between junction 1 and junction 5 in configuration I is $50°$C.	**(D)** The steady-state heat flux in configuration I is $200{,}000$ W/m^2.

Answer: A, B, D

Q.25 A $0.1\ \mathrm{kg}$ bullet traveling horizontally at $200\ \mathrm{m/s}$ gets embedded in a $1.9\ \mathrm{kg}$ stationary block. The block rests on a frictionless horizontal surface a short distance away from an unattached, ideal horizontal spring of force constant $200\ \mathrm{N/m}$ fixed to a wall. After the bullet embeds, the combined mass slides and compresses the spring until it momentarily stops. Which of the following statements is/are correct?



(A) The velocity of the combined mass immediately after the initial collision is 10 m/s.	**(B)** The loss of kinetic energy during the bullet-block collision is 1900 J.
(C) The maximum compression of the spring is 1 m.	**(D)** The magnitude of the impulse exerted by the spring on the combined mass from initial contact until it momentarily stops is 40 N·s.

Answer: A, B, C

PHYSICS SECTION 3 (Maximum Marks : 20)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

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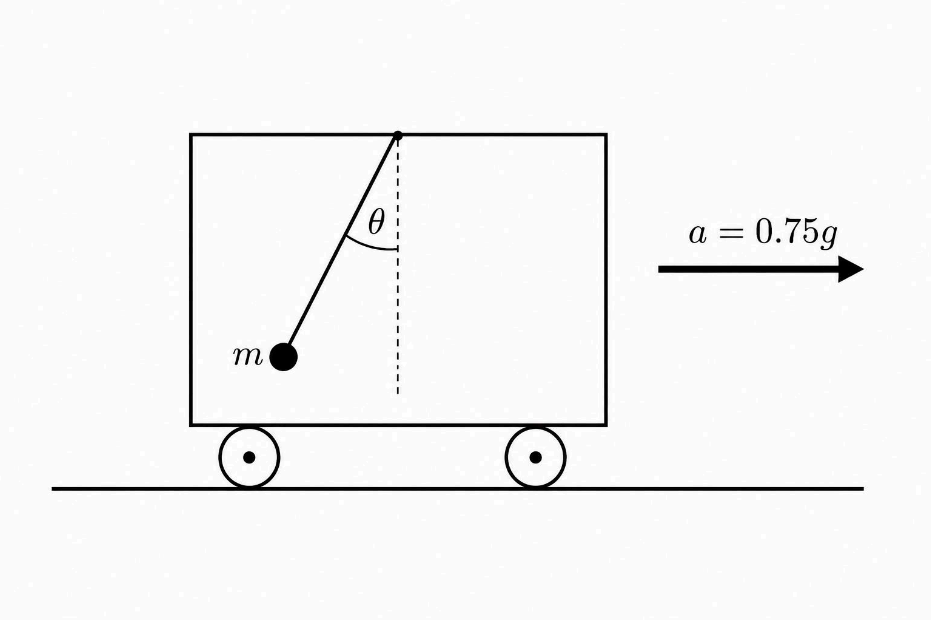
Zero Marks : 0 If no numerical answer is given (i.e. the question is unanswered);
Negative Marks : -1 In all other cases

Q.26 A hollow spherical shell of mass M and radius R is completely filled with a non-viscous liquid of mass M. The sphere rolls without slipping on a horizontal surface with a linear center-of-mass velocity v. The ratio of the system's total kinetic energy to its translational kinetic energy is given as α. The value of α is ____. (Round off to two decimal places)

Answer: [1.3 to 1.4]

Q.27 A simple pendulum consisting of a bob of mass m and a massless string of length L is suspended from the roof of a cart. Initially, the cart moves horizontally with a constant acceleration of $0.75g$, and the pendulum rests in its equilibrium position relative to the cart. Instantaneously, the cart's acceleration is reversed in direction while maintaining the exact same magnitude, without altering the cart's instantaneous velocity. The pendulum bob begins to swing within the cart.

Considering no energy dissipation and evaluating the motion in the cart's frame, the maximum tension developed in the string during the subsequent swing is T_{max}. The exact value of the ratio $\dfrac{T_{max}}{mg}$ is ____.



Answer: [3 to 3.1]

Q.28 A sonometer wire of constant linear mass density $\mu = 0.01\ \mathrm{kg/m}$ is maintained continuously in its 2^{nd} overtone. Initially, the wire is under a tension of $16\ \mathrm{N}$ and vibrates at a frequency of $60\ \mathrm{Hz}$.An experimenter performs two consecutive adjustments to the system:

Adjustment 1: The tension in the wire is increased to $64\ \mathrm{N}$ while its vibrating length is held strictly constant.

Adjustment 2: The tension is then further increased to $144\ \mathrm{N}$. During this specific adjustment, the experimental setup is manipulated such that the frequency of vibration is maintained directly proportional to the applied tension.

Determine the final vibrating length of the wire (in meters) after the second adjustment is complete. (Round off to two decimal places).

Answer: [0.6 to 0.7]

Q.29 An oppositely charged point charge undergoes small harmonic oscillations along the axis of a fixed, uniformly charged ring of radius r_1 with a time period of $24\ \mathrm{ms}$. A second identical point charge also undergoes small harmonic oscillations along the axis of another fixed ring of equal total charge but radius r_2, such that $r_1 = 1.21 r_2$. If both charges start their oscillations in phase, the time interval between consecutive instances when they are again exactly in phase with each other is $\frac{24}{p}$ ms. The value of p is ____.

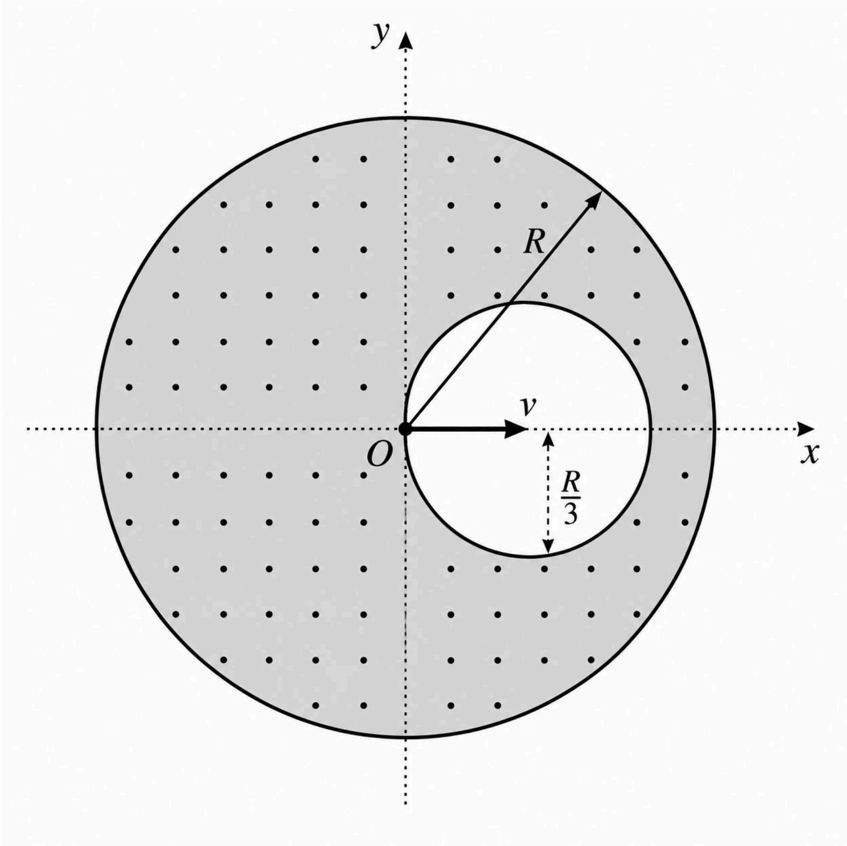
(If the numerical value has more than two decimal places, truncate the value to TWO decimal places.)

Answer: [0.3 to 0.4]

Q.30 A very long, solid, non-magnetic cylindrical conductor of radius $12\ \mathrm{cm}$ has a cylindrical cavity of radius $4\ \mathrm{cm}$ bored through its entire length. The axis of the main cylinder coincides with the z-axis, while the axis of the cavity is parallel to the z-axis and intersects the xy-plane at the coordinate $(4\ \mathrm{cm}, 0, 0)$. The remaining solid portion of the conductor carries a steady total current of $1280A$ directed along the positive z-axis, distributed such that the current density is uniform across the solid cross-section.

An electron gun is placed at the origin $(0, 0, 0)$ and projects an electron directly along the positive x-axis into the cavity region. The electron travels in a curved trajectory and just barely grazes the far wall of the cavity. Assuming no other magnetic or electric fields are present outside of those generated by the cylinder, determine the initial kinetic energy of the projected electron in eV.

(Take the specific charge of an electron as $e/m = 1.8 \times 10^{11}\ \mathrm{C/kg}$. Express your answer as a decimal number rounded to two decimal places.)



Answer: [366 to 370]

PHYSICS SECTION 4 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

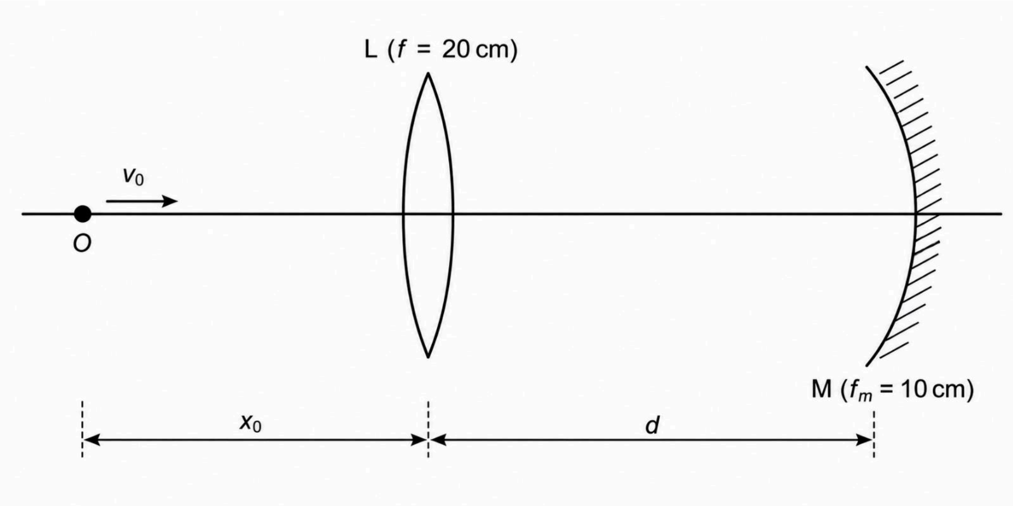
  Full Marks : +3 If **ONLY** the numerical answer is in acceptable range

  Zero Marks : 0 In all other cases

Q.31 **Paragraph:** A point object O moves along the principal axis of a coaxial optical system comprising a thin converging lens L ($f = 20$ cm) and a concave mirror M ($f_m = 10$ cm), separated by a distance $d = 50$ cm.

Initially at time $t = 0$, O is located $x_0 = 60$ cm to the left of L and approaches it with a uniform speed $v_0 = 5$ cm/s. Light from O undergoes refraction at L, reflection at M, and a second refraction at L to form a final instantaneous image I.

Let $v(t)$ be the speed of this final image, and $D(t)$ be its absolute distance from the optical center of the lens at any given time t. All calculations consider paraxial rays.



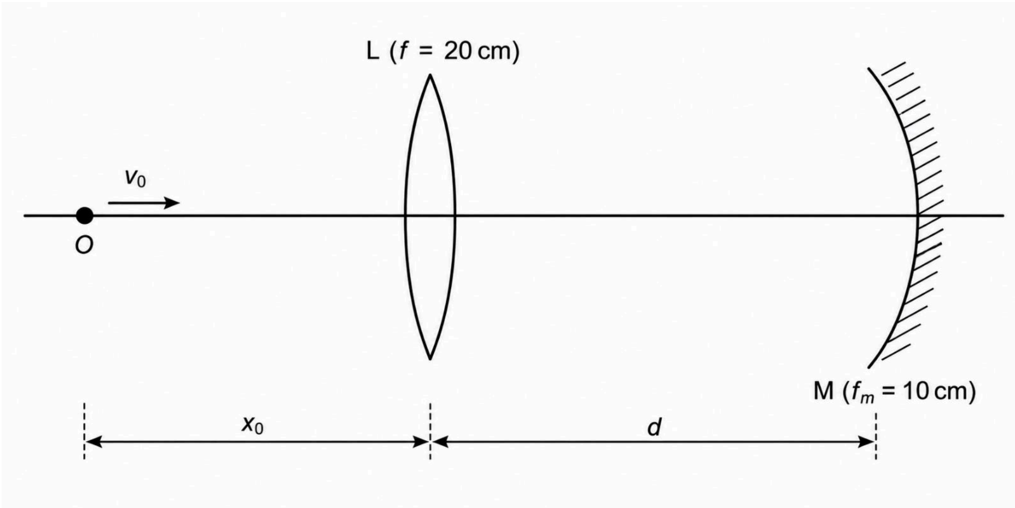
Question: What is the value of the ratio $\dfrac{v(6)}{v(0)}$? (Round off to two decimal places)

Answer: 0.64

Q.32 **Paragraph:** A point object O moves along the principal axis of a coaxial optical system comprising a thin converging lens L ($f = 20$ cm) and a concave mirror M ($f_m = 10$ cm), separated by a distance $d = 50$ cm.

Initially at time $t = 0$, O is located $x_0 = 60$ cm to the left of L and approaches it with a uniform speed $v_0 = 5$ cm/s. Light from O undergoes refraction at L, reflection at M, and a second refraction at L to form a final instantaneous image I.

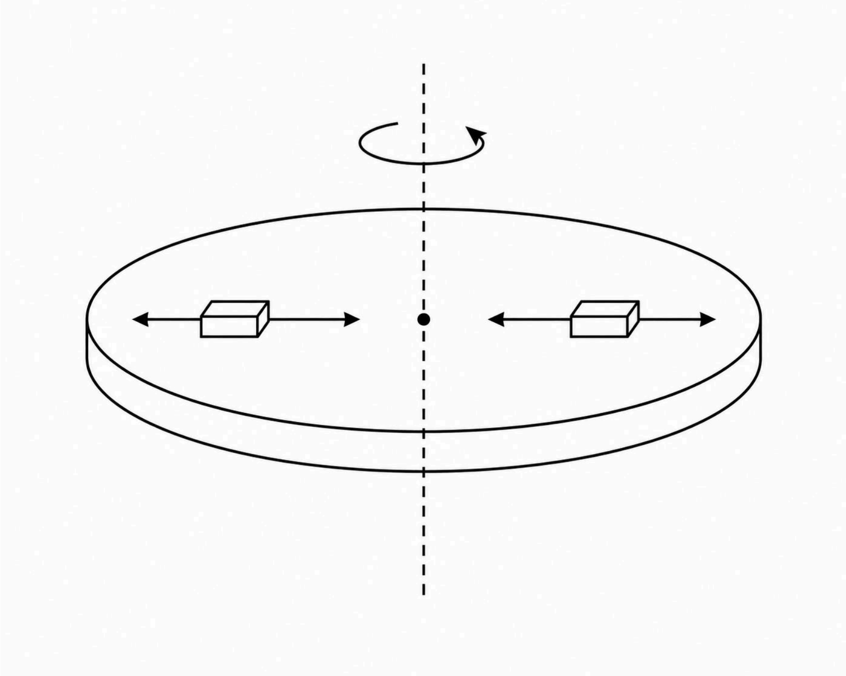
Let $v(t)$ be the speed of this final image, and $D(t)$ be its absolute distance from the optical center of the lens at any given time t. All calculations consider paraxial rays.



Question: What is the value of the ratio $\dfrac{D(6)}{D(0)}$? (Round off to two decimal places)

Answer: 0.6

Q.33 **Paragraph:** A horizontal circular platform is rotating freely about a fixed, frictionless vertical axis passing through its center. Due to a system of internally driven radial masses, the moment of inertia of the entire system varies with time t as $I = (0.5 + 0.05 \sin \pi t)\ \mathrm{kg} \cdot \mathrm{m}^2$. The system is initially set in motion such that its constant angular momentum is $3\ \mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}$. No external torques act on the system during the motion. Consequently, the rotational kinetic energy and angular speed continually fluctuate.

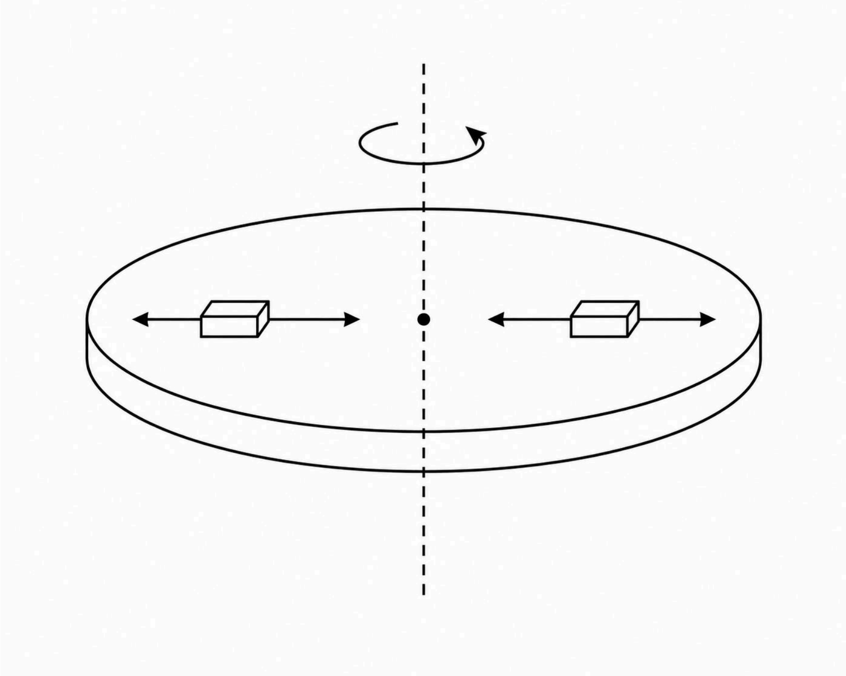


Question: The rotational kinetic energy of the system fluctuates between a maximum and a minimum value. The difference between these two extreme kinetic energies, in Joules (J), is _____. (Round off to two decimal places)

Answer: [1.8 to 1.82]

Q.34

Paragraph: A horizontal circular platform is rotating freely about a fixed, frictionless vertical axis passing through its center. Due to a system of internally driven radial masses, the moment of inertia of the entire system varies with time t as $I = (0.5 + 0.05 \sin \pi t)\ \mathrm{kg} \cdot \mathrm{m}^2$. The system is initially set in motion such that its constant angular momentum is $3\ \mathrm{kg} \cdot \mathrm{m}^2/\mathrm{s}$. No external torques act on the system during the motion. Consequently, the rotational kinetic energy and angular speed continually fluctuate.



Question: The magnitude of the rate of change of the system's rotational kinetic energy (in J/s) at the instant the moment of inertia first passes through its mean value of $0.5\ \mathrm{kg} \cdot \mathrm{m}^2$ after $t = 0$ is _____. (Round off to two decimal places)

Answer: [2.81 to 2.83]

CHEMISTRY SECTION 1 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR options** (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks : +3 If **ONLY** the correct option is chosen
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : -1 In all other cases.

Q.35 Identify the primary therapeutic classifications for Equanil, Aspirin, and Norethindrone, sequentially.

(A) $[\text{Tranquilizer}] \rightarrow [\text{Non-narcotic analgesic}] \rightarrow [\text{Antifertility}]$	**(B)** $[\text{Tranquilizer}] \rightarrow [\text{Narcotic analgesic}] \rightarrow [\text{Antifertility}]$
(C) $[\text{Antihistamine}] \rightarrow [\text{Non-narcotic analgesic}] \rightarrow [\text{Antibiotic}]$	**(D)** $[\text{Antidepressant}] \rightarrow [\text{Narcotic analgesic}] \rightarrow [\text{Antifertility}]$

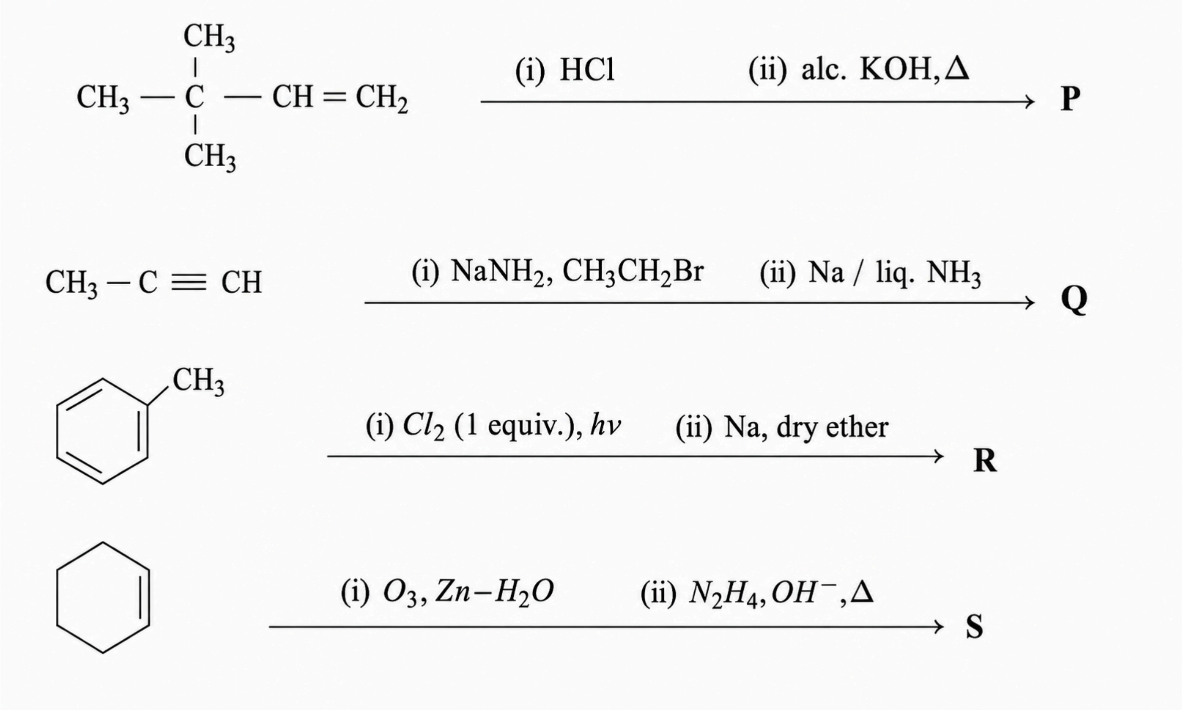
Answer: A

Q.36 The reaction sequence that produces a primary aromatic amine as the final major product is:

(A) $\text{Benzamide} \xrightarrow{\text{(i) } LiAlH_4\text{,ether} \quad \text{(ii) } H_2O}$	**(B)** $\text{Nitrobenzene} \xrightarrow{\text{(i) } Zn/NH_4Cl \quad \text{(ii) } NaOI}$
(C) $\text{Phthalimide} \xrightarrow{\text{(i) } KOH\text{,}EtOH \quad \text{(ii) Chlorobenzene,}\Delta \quad \text{(iii) } N_2H_4}$	**(D)** $\text{Benzamide} \xrightarrow{Br_2\text{,}KOH\text{,}\Delta}$

Answer: D

Q.37 Determine the major products $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{R}$, and $\mathbf{S}$ formed in the following reaction sequences.



Evaluate the total number of sp^3 hybridized carbon atoms present in each of the final major products. Which of the following statements is correct?

(A) The sum of sp^3 hybridized carbons in $\mathbf{Q}$ and $\mathbf{R}$ is exactly equal to the number of sp^3 hybridized carbons in $\mathbf{S}$.	**(B)** $\mathbf{P}$ is capable of showing geometrical isomerism and contains four sp^3 hybridized carbons.
(C) The difference in the number of sp^3 hybridized carbons between $\mathbf{S}$ and $\mathbf{P}$ is equal to the number of sp^3 hybridized carbons in $\mathbf{R}$.	**(D)** $\mathbf{Q}$ is the *cis*-isomer and contains three sp^3 hybridized carbons.

Answer: C

Q.38 During Lassaigne's test for nitrogen, the characteristic Prussian blue precipitate formed is a coordination complex with the specific formula:

(A) $Fe_4[Fe(CN)_6]_3$	**(B)** $Fe_3[Fe(CN)_6]_4$
(C) $Fe_2[Fe(CN)_6]_3$	**(D)** $Fe_4[Fe(CN)_6]_2$

Answer: A

CHEMISTRY SECTION 2 (Maximum Marks : 16)

- This section contains **FOUR (04)** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s)
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 **ONLY** if (all) the correct option(s) is(are) chosen;
  Partial Marks : +3 If all the four options are correct but **ONLY** three options are chosen;
  Partial Marks : +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;
  Partial Marks : +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;
  Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
  Negative Marks : -2 In all other cases.

Q.39 The correct statement(s) regarding the chemical and physical properties of p-block elements is(are)

(A) The spontaneous acidic dissociation of hydrohalic acids in aqueous media increases from HF to HI primarily because the increasing electronegativity of the heavier halogens enhances the $H - X$ bond polarity.	**(B)** The basic character of Group 15 trihydrides decreases progressively from NH_3 to BiH_3 as the non-bonding electron pair becomes more diffuse over a larger atomic volume.
(C) The electron gain enthalpy of oxygen is distinctly less negative than that of sulfur owing to substantial interelectronic repulsions within the highly compact $2p$ subshell of the oxygen atom.	**(D)** The complete hydrolysis of xenon hexafluoride (XeF_6) yields highly explosive xenon trioxide (XeO_3) through a disproportionation reaction where xenon undergoes simultaneous oxidation and reduction.

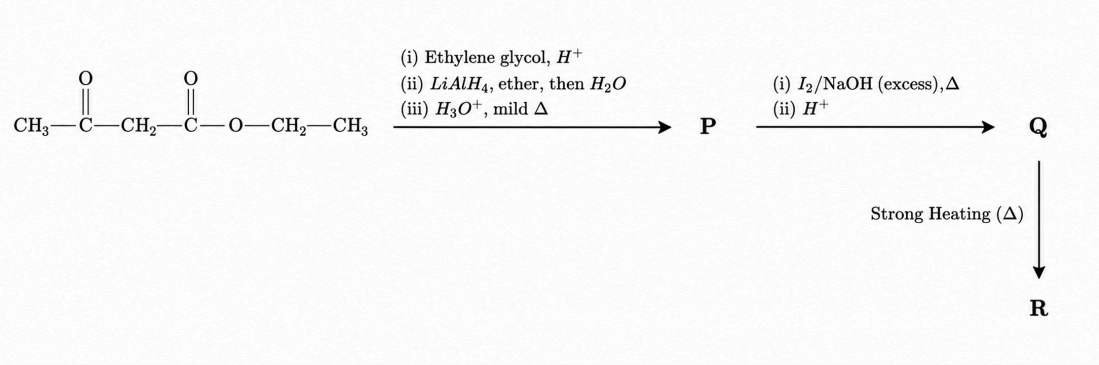
Answer: B, C

Q.40 Consider the thermodynamic and electrochemical principles governing the extraction of metals. Which of the following statements is/are CORRECT?

(A) In an Ellingham diagram, the sudden positive increase in the slope of the ΔG° vs T curve for the formation of a metal oxide (e.g., $Mg \to MgO$ or $Zn \to ZnO$) indicates a phase change (melting or boiling) of the metal itself, leading to a decrease in the magnitude of entropy change (ΔS°) for the oxidation reaction.	**(B)** According to Ellingham diagrams, a metal will thermodynamically reduce the oxide of any other metal that lies above it in the diagram at a specific temperature.
(C) In the Macarthur-Forrest cyanide process for the extraction of Silver, the displacement of Ag from the $[Ag(CN)_2]^-$ complex by Zinc is thermodynamically feasible primarily because Zinc has a much higher (more positive) standard reduction potential (E°) than Silver.	**(D)** Carbon reduction of ZnO to Zn (smelting) becomes thermodynamically feasible only above the temperature where the ΔG° vs T line for the reaction $2C + O_2 \to 2CO$ crosses below the line for $2Zn + O_2 \to 2ZnO$.

Answer: B, D

Q.41 For the reaction sequence given below, the correct statement(s) is(are):



(A) $\mathbf{P}$ gives a positive 2,4-DNP test but fails to give a silver mirror with Tollen's reagent.	**(B)** $\mathbf{R}$ is an α, β-unsaturated carboxylic acid that can exhibit geometrical isomerism.
(C) The formation of $\mathbf{Q}$ from $\mathbf{P}$ involves an enolate (carbanion) intermediate and produces a yellow precipitate.	**(D)** $\mathbf{Q}$ upon mild heating undergoes intermolecular esterification to form a cyclic diester known as a lactide.

Answer: A, C

Q.42 Consider the following sequence for two pure volatile liquids, Acetone ($\mathbf{X}$) and Chloroform ($\mathbf{Y}$):

$$\mathbf{X} + \mathbf{Y} \xrightarrow{\text{mix at 298 K}} \textbf{Solution P} \text{ (non-ideal)}$$

$$\mathbf{P} \xrightarrow{\text{Careful Fractional Distillation}} \textbf{First Distillate Q} + \textbf{Final Residue R}$$

Which of the following statement(s) is(are) correct?

(A) ΔS_{mix} for the formation of $\mathbf{P}$ is negative.	**(B)** $\mathbf{P}$ can have a vapor pressure lower than pure $\mathbf{Y}$.
(C) $\mathbf{R}$ is a constant-boiling mixture at a maximum temperature.	**(D)** $\mathbf{Q}$ is an azeotropic mixture of $\mathbf{X}$ and $\mathbf{Y}$.

Answer: B, C

CHEMISTRY SECTION 3 (Maximum Marks : 20)

- This section contains **FIVE (05)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +4 If **ONLY** the numerical answer is in acceptable range
  Zero Marks : 0 If no numerical answer is given (i.e. the question is unanswered);
  Negative Marks : -1 In all other cases

Q.43 The standard bond dissociation enthalpy (ΔH°) of a homonuclear diatomic gas $A_2(g)$ is exactly equal to the energy of one mole of photons having a wavelength of 400 nm.

For the dissociation reaction $A_2(g) \rightleftharpoons 2A(g)$, the standard entropy change ΔS° is 99 $\text{J K}^{-1} \text{ mol}^{-1}$. (Assume both ΔH° and ΔS° are independent of temperature).

Initially, 1 mole of $A_2(g)$ is taken in an evacuated rigid container of volume V. The container is heated to a temperature T, where the system establishes equilibrium. At this specific temperature T, the standard Gibbs free energy change (ΔG°) for the reaction is zero.

If the degree of dissociation (α) of $A_2(g)$ at this equilibrium state is 0.5, calculate the initial volume of the container V (in Liters).

(Use the following constants: $h = 6.6 \times 10^{-34} \text{ J s}$, $c = 3 \times 10^8 \text{ m s}^{-1}$, $N_A = 6 \times 10^{23} \text{ mol}^{-1}$, $R = 8.3 \text{ J K}^{-1} \text{ mol}^{-1}$ and $R = 0.083 \text{ L bar K}^{-1} \text{ mol}^{-1}$. Standard state pressure $P^\circ = 1 \text{ bar}$)

Answer: 498

Q.44 Consider the gas-phase decomposition of reactant X through two parallel, first-order pathways:

Pathway 1: $X(g) \longrightarrow Y(g) + Z(g)$ (Activation energy $= E_{a1}$)
Pathway 2: $X(g) \longrightarrow W(g)$ (Activation energy $= E_{a2}$)

When 1.0 mol of $X(g)$ is allowed to decompose completely in a closed vessel at 300 K, the final reaction mixture contains 0.2 mol of $W(g)$.

However, when the exact same 1.0 mol sample is allowed to decompose completely at 600 K, the final reaction mixture contains 0.5 mol of $W(g)$.

Assuming the pre-exponential factors (A_1 and A_2) for both pathways are independent of temperature, calculate the value of $\dfrac{E_{a2} - E_{a1}}{2.303R}$ (in Kelvin). Round off to two decimal places.

(Use: $\log_{10} 2 = 0.301$)

Answer: 361.2

Q.45 The ratio (R) of the concentration of a complex to free metal ions in an aqueous solution obeys the overall stability relation $R = \beta_n [L]^n$. At a given temperature, from $10 \mathrm{~mM}$ and $16 \mathrm{~mM}$ free ligand concentrations ($[L]$), the ratios R are measured to be 4 and 10.24, respectively. At this temperature, the ratio R from a $20 \mathrm{~mM}$ free ligand concentration will be ______.

Use: $\log_{10} 2 = 0.3$

Answer: 16

Q.46 Consider a defect recombination process in a crystal lattice where the rate is measured to be $k[N_i][N_v]$. At the start of the process, the vacancy concentration, $[N_v]_0$, is 10-times the interstitial concentration, $[N_i]_0$. The process can be considered to be a pseudo first order reaction with assumption that $k[N_v] = k'$ is constant. Due to this assumption, the relative error (in %) in the rate when this annihilation is 40% complete, is ____. (Round off to two decimal places)

[Let k and k' represent corresponding rate constants]

Answer: [4.1 to 4.2]

Q.47 An optically active alkene C_6H_{12} hydrogenates to an achiral alkane. Its anti-Markovnikov hydrobromination yields compound Y. Find the ratio of Y's molar mass to the maximum number of theoretical stereoisomers possible for the structure of Y. (Round off to 2 decimal places. $\mathrm{C} = 12$, $\mathrm{H} = 1$, $\mathrm{Br} = 80$).

Answer: 82.5

CHEMISTRY SECTION 4 (Maximum Marks : 12)

- This section contains **FOUR (04)** questions.
- The answer to each question is a **NUMERICAL VALUE.**
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated **according to the following marking scheme:**

  Full Marks : +3 If **ONLY** the numerical answer is in acceptable range

  Zero Marks : 0 In all other cases

Q.48

Paragraph: An optically active organic compound $\mathbf{P}$ ($C_{11}H_{14}O_2$) undergoes complete acidic hydrolysis to yield two organic compounds, $\mathbf{Q}$ and $\mathbf{R}$. Compound $\mathbf{R}$ is optically active and yields a yellow precipitate upon warming with alkaline iodine. Compound $\mathbf{Q}$ does not decolorize bromine water but undergoes electrophilic aromatic substitution. Treatment of exactly 15.25 g of $\mathbf{Q}$ with excess $Br_2/FeBr_3$ yields a single major monobrominated product $\mathbf{S}$ with a 67% reaction yield. Dehydration of 0.50 moles of $\mathbf{R}$ with concentrated H_2SO_4 produces $\mathbf{T}$ as the major product, which exhibits geometrical isomerism. Complete ozonolysis of the formed $\mathbf{T}$ followed by reductive workup gives exclusively organic molecule $\mathbf{U}$. The entire quantity of $\mathbf{U}$ is heated with Tollens' reagent to deposit a silver mirror, proceeding with an 81.3% overall sequential efficiency from $\mathbf{R}$.

[Given atomic masses: H = 1, C = 12, O = 16, Br = 80, Ag = 108]

Question: The mass of $\mathbf{S}$ formed (in grams) is ____. (Round off to two decimal places)

Answer: [16 to 18]

Q.49

Paragraph: An optically active organic compound $\mathbf{P}$ ($C_{11}H_{14}O_2$) undergoes complete acidic hydrolysis to yield two organic compounds, $\mathbf{Q}$ and $\mathbf{R}$. Compound $\mathbf{R}$ is optically active and yields a yellow precipitate upon warming with alkaline iodine. Compound $\mathbf{Q}$ does not decolorize bromine water but undergoes electrophilic aromatic substitution. Treatment of exactly 15.25 g of $\mathbf{Q}$ with excess $Br_2/FeBr_3$ yields a single major monobrominated product $\mathbf{S}$ with a 67% reaction yield. Dehydration of 0.50 moles of $\mathbf{R}$ with concentrated H_2SO_4 produces $\mathbf{T}$ as the major product, which exhibits geometrical isomerism. Complete ozonolysis of the formed $\mathbf{T}$ followed by reductive workup gives exclusively organic molecule $\mathbf{U}$. The entire quantity of $\mathbf{U}$ is heated with Tollens' reagent to deposit a silver mirror, proceeding with an 81.3% overall sequential efficiency from $\mathbf{R}$.

[Given atomic masses: H = 1, C = 12, O = 16, Br = 80, Ag = 108]

Question: The mass of the silver mirror deposited (in grams) is ____. (Round off to two decimal places)

Answer: [174 to 176.5]

Q.50

Paragraph: Exactly 100 mL of a 0.20 M aqueous solution of a weak monoprotic acid HX ($K_a = 2.0 \times 10^{-5}$) is mixed completely with 100 mL of 0.10 M NaOH to generate a buffered solution $\mathbf{P}$. Subsequently, a highly soluble metallic salt $M(NO_3)_2$ is slowly introduced into $\mathbf{P}$. This addition continues until the sparingly soluble salt MX_2 ($K_{sp} = 3.2 \times 10^{-11}$) just begins to precipitate, establishing equilibrium state $\mathbf{Q}$.

Question: The pH of the buffered solution $\mathbf{P}$ at standard temperature is ____. (Round off to two decimal places)

(Assume $\log 2 = 0.30$ and that addition of the solid salt does not change the volume)

Answer: 4.7

Q.51

Paragraph: Exactly 100 mL of a 0.20 M aqueous solution of a weak monoprotic acid HX ($K_a = 2.0 \times 10^{-5}$) is mixed completely with 100 mL of 0.10 M NaOH to generate a buffered solution $\mathbf{P}$. Subsequently, a highly soluble metallic salt $M(NO_3)_2$ is slowly introduced into $\mathbf{P}$. This addition continues until the sparingly soluble salt MX_2 ($K_{sp} = 3.2 \times 10^{-11}$) just begins to precipitate, establishing equilibrium state $\mathbf{Q}$.

Question: The minimum concentration of M^{2+} ions required in state $\mathbf{Q}$ to initiate precipitation is ____ $\times 10^{-8}$ M.

(Assume $\log 2 = 0.30$ and that addition of the solid salt does not change the volume. If the numerical value filling the blank has more than two decimal places, round it off to two.)

Answer: [1.2 to 1.3]